

Matrix Inverse Modulo 26 in Hill Cipher

Introduction

In the Hill Cipher, encryption is performed using a key matrix K over $\mathbb{Z}_{26}$. The plaintext vector P is multiplied with K modulo 26:

$$C \equiv P \cdot K \pmod{26}$$

where C is the ciphertext. For decryption, we require the modular inverse of K , denoted K^{-1} , so that:

$$P \equiv C \cdot K^{-1} \pmod{26}$$

Thus, the computation of the matrix inverse modulo 26 is essential for decryption.

Matrix Inverse in Modular Arithmetic

Step 1: Determinant

The determinant of the key matrix K is computed in the usual way. For a 2×2 matrix:

$$K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(K) = ad - bc$$

For a 3×3 matrix:

$$\det(K) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

where

$$K = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Step 2: Modular Inverse of Determinant

Since calculations are modulo 26, we must ensure:

$$\gcd(\det(K), 26) = 1$$

Only then does $\det(K)$ have a multiplicative inverse modulo 26. This value, denoted $\det(K)^{-1}$, is found using the Extended Euclidean Algorithm.

Step 3: Adjoint Matrix

The adjoint (or adjugate) matrix is the transpose of the cofactor matrix.

Case 1: 2×2 Matrix

$$\text{adj}(K) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Case 2: 3×3 Matrix For each entry (i, j) of K , the cofactor is computed as:

$$\text{Cofactor}_{ij} = (-1)^{i+j} \cdot M_{ij}$$

where M_{ij} is the minor determinant obtained by removing row i and column j .
For example, for position $(0, 0)$:

$$M_{00} = \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} = ei - fh$$

The full cofactor matrix is then transposed to obtain $\text{adj}(K)$.

Step 4: Inverse Matrix Modulo 26

Finally, the inverse matrix is computed as:

$$K^{-1} \equiv \det(K)^{-1} \cdot \text{adj}(K) \pmod{26}$$

This means each entry of the adjoint is multiplied by the modular inverse of the determinant, and then reduced modulo 26.

Decryption

Once K^{-1} is found, decryption is performed as:

$$P \equiv C \cdot K^{-1} \pmod{26}$$

This recovers the original plaintext vector from the ciphertext.

Conclusion

The Hill Cipher requires matrix inversion in modular arithmetic for decryption. For a 2×2 or 3×3 key, the process involves:

1. Calculating the determinant,
2. Finding its modular inverse modulo 26,
3. Computing the adjoint matrix,
4. Multiplying the adjoint by the modular determinant inverse modulo 26.

This ensures that the ciphertext can be correctly decrypted back to the original plaintext.